

CSCI 332, Fall 2026

Homework 4

Due Monday, September 21 Anywhere on Earth (6am Tuesday)

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Gradescope.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.
- Use Gradescope to assign problems to the correct page(s) in your solution. If you do not do this correctly, we will ask you to resubmit.

Academic Integrity

Remember, you may access **any** resource in preparing your solution to the homework. However, you **must**

- write your solutions in your own words, and
- credit every resource you use (for example: “Bob Smith helped me on this problem. He took this course at UM in Fall 2020”; “I found a solution to a problem similar to this one in the lecture notes for a different course, found at this link: www.profzeno.com/agreatclass/lecture10”; “Here is a link to the full transcript of my chat with ChatGPT about the problems on this HW”.) Ask if you need any clarifications about how to properly cite your sources!

1. (0 points) List the resources you used for each problem in this homework. If you used no resources except lecture notes and videos and the linked textbook, you can say “none”. If you used an AI tool, you should put a link to your full conversation here, and you must use the [provided prompt](#) to guide the AI to act as a tutor, not an answer-provider.

You will be asked to resubmit your homework if this section is blank.

Remember that no matter what resources you consulted, you must write your solutions *in your own words*.

2. (1 point) (without 3, 2 points) Write the recurrence describing the following algorithm’s runtime. (That is, fill in the blanks below.)

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Alg(array A of length n):
  If n = 1:
    Return A
  Else:
    Return Alg(array formed by taking the values at odd indices of A)
  
```

$$T(n) = \begin{cases} \underline{\hspace{4cm}} & \text{if } n \underline{\hspace{1cm}} \\ \underline{\hspace{4cm}} & \text{if } n \underline{\hspace{1cm}} \end{cases}$$

3. (3 points) Using induction, prove that the sum of the degrees of all vertices in an undirected graph is equal to two times the number of edges in the graph. To earn any points, your proof must use the exact form we learned in class on Friday (universal declaration, inductive hypothesis, cases, wrap-up statement.)
4. (6 points) (without (b), 8 points) Suppose you are given a stack of n pancakes of different sizes. You want to sort the pancakes so that smaller pancakes are on top of larger pancakes. The only operation you can perform is a *flip*—insert a spatula under the top k pancakes, for some integer k between 1 and n , and flip them all over.

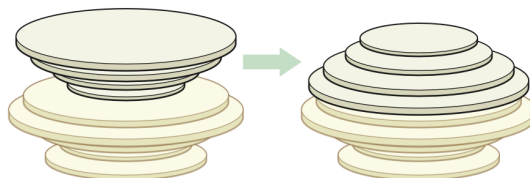


Figure 1. Flipping the top four pancakes.

- (a) Let *PancakeSort* be an algorithm that sorts a stack of pancakes. Given a stack of n pancakes S , describe how you could perform a constant number of flips on S so that calling *PancakeSort* on the top $n - 1$ pancakes completes the sorting of S .
- (b) Write the full recursive *PancakeSort* algorithm.
- (c) Write the recurrence for your *PancakeSort* algorithm’s runtime $T(n)$.